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Paleomagnetic evidence for the Miocene counter-clockwise rotation of Northeast Japan—rifting process of the Japan Arc

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Paleomagnetic results have been obtained from more than 700 oriented samples in igneous rocks of Cretaceous to Miocene age from Northeast Japan. The remanent magnetizations of welded tuffs with age between 32 Ma and 21 Ma old from 17 widely distributed sampling sites in Northeast Japan are fairly well grouped with a mean of $D = -41.2°$, $I = 56.5°$, $\alpha_{95} = 7.2°$. The remanent magnetization of welded tuffs of 14 Ma and 11 Ma from 4 sites shows a northerly direction ($D = -11.7°$, $I = 60.5°$, $\alpha_{95} = 15.3°$). These data imply that Northeast Japan was apparently subjected to a counter-clockwise rotation through $47 \pm 14°$ about a vertical axis on the representative point of Northeast Japan (141°E, 39°N) during the past 21 Ma. Comparing with the period of the clockwise rotation of Southwest Japan, the differential rotation between Northeast Japan and Southwest Japan is concluded to have occurred concurrently during the period between 21 Ma and 12 Ma. A rifting model of the Japan arc is proposed: Northeast Japan rotated counter-clockwise through 50° about a northern pivot at 146°E, 44°N, while Southwest Japan rotated clockwise through 54° about a southern pivot at 129°E, 34°N. The differential rotation about different rotation pivots gives rise to the opening of a diamond-shaped back arc basin, the Japan Sea.

1. Introduction

The morphological features of the Japan Trench–Japan arc–Japan Sea are typical of trench-arc-basin systems in the western Pacific. The origin of some basins in these systems (e.g. Shikoku, Parece Vela, Mariana and South China basins) is ascribed to sea floor spreading behind island arcs [1–5]. The Japan Sea has been also explained as being caused by back arc spreading, on the basis of the geological and geophysical evidence obtained from its basin [6–9]. Consequently, the Japan arc was suggested to be a continental fragment rifted from the Asian continental margin due to crustal extension during spreading of the Japan Sea [10,11]. The Japan arc–Japan Sea system provides an opportunity to study the geodynamics of the rifting of the continental fragment associated with back arc spreading.

The evidence for the rifting phenomena of the Japan arc is preserved in the igneous materials on the Japanese islands as well as in the sea floor of the Japan Sea. The intent of this paper is to present an outline of the rifting process from the viewpoint of the paleomagnetism of the Japanese islands.

The clue to the rifting process of the Japan arc is the tectonic event of differential rotational motion of Northeast and Southwest Japan arcs since the Cretaceous period. Previous paleomagnetic studies on Cretaceous to Paleogene rocks [12–14] demonstrated that the Northeast Japan arc rotated counter-clockwise through more than 30° and the Southwest Japan arc was subjected to clockwise rotation through about 30° with respect to the Eurasia. The rotational motion of both arcs has been believed to have ceased by Late Cretaceous, because the paleomagnetic data from Northeast Japan showed that the declination change took place at about 80 Ma from westward to eastward [12]. In recent years Southwest Japan has been the focus of several paleomagnetic investigations [15–18]: Improvement of paleomagnetic data suggests that the rotation of Southwest Japan relative to the Eurasia started later than 21 Ma. This result gives rise to the hypothesis on the Miocene origin

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of rifting of the Japan arc.

The present investigation is focused on the Northeast Japan arc: The study was undertaken to discover the paleomagnetic evidence for the Miocene counter-clockwise rotation of the Northeast Japan arc in an attempt to determine the timing of rifting of the Japan arc.

2. Geology and sampling

Four different stages can be distinguished for igneous activity in Northeast Japan since the Cretaceous [19,20]. Granites and granodiorites of Cretaceous to early Paleogene age, felsic igneous rocks of Paleogene age, the "Green Tuff" of late Tertiary age, and andesitic materials of late Miocene to the Present resulted from four respective stages of activity (see Fig. 1). Intrusion of plutons commenced in the late stage of Early Cretaceous throughout the entire region of Northeast Japan, e.g., Kitakami, Abukuma, Ashio and Iide-Asahi. The plutonic activity took place at first in the Pacific coast side at 110–120 Ma and the active area subsequently shifted toward the Japan Sea side. In the early Paleogene period (50–65 Ma), the plutonic rocks accompanied with the felsic volcanic rocks occurred in the Uetsu area which extends from Tsukuba, northward through Asahi to Oga Peninsula. Intensive volcanism was reactivated in the western side of Northeast Japan since late Oligocene. Their volcanic products called "Green Tuff" are dominated by a series of pyroclastic rocks, in which some constituents are altered to chlorite, epidote, sericite and zeolite. The green tuff volcanism was initiated by voluminous intermediate and acidic volcanic rocks in the subaerial environment, and followed by lesser amounts of submarine basalts and andesites. The andesitic magma erupted over the Ou Mountain Range in the central part of Northeast Japan during late Miocene to the Present.

Our paleomagnetic approach is to analyze the welded tuffs of Paleogene to Miocene age, because a reliable remanent magnetization could be expected to be obtained [16,21,22] and tilting of the welded tuff could be corrected on the basis of the eutaxitic structure (lineation of stretched pumice and aligned phenocrysts) [23]. Since the eutaxitic structure does not always indicate the paleohorizontal, sampling in each welded tuff was carried

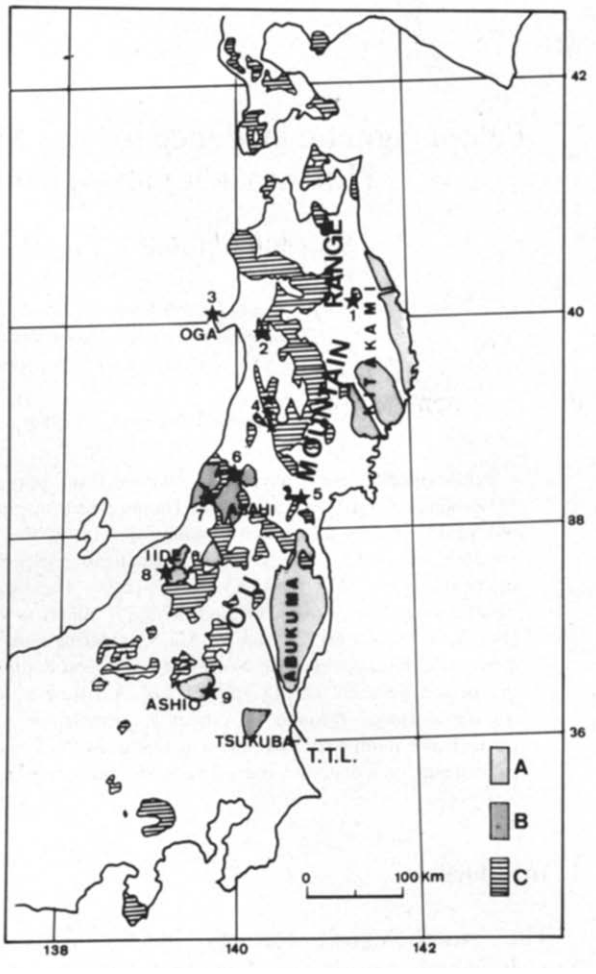


Fig. 1. Locations of paleomagnetic sampling areas (shown by stars) in Northeast Japan together with distribution of Cretaceous granitic rocks (*A*), Paleogene granitic rocks (*B*), and Miocene volcanic rocks (*C*): *1* = Ninohe, *2* = Taiheizan, *3* = Oga Peninsula, *4* = Yuzawa, *5* = Zao, *6* = Northern Asahi, *7* = Western Asahi, *8* = Iide and *9* = Ashio. · *T.T.L.* = Tanakura Tectonic Line.

out at widely spaced outcrops for reducing the error caused by uncertainty in tectonic correction. Samples were also collected from plutonic rocks of late Cretaceous and early Paleogene when few welded tuffs had been produced. More than 700 samples were collected from nine areas (Fig 1); (1) Ninohe, (2) Taiheizan, (3) Oga Peninsula, (4) Yuzawa, (5) Zao, (6) Northern Asahi, (7) Western Asahi, (8) Iide and (9) Ashio. These areas have been geologically mapped in detail [24–33]. Sampling was mostly from rock units whose ages had

been calculated by either K-Ar or fission track dating [34–38]. Sampling localities are listed in Table 1.

Samples were collected by hand sampling; a magnetic compass was used for orientation. Each site typically comprised ten independently oriented samples distributed over distance ranging up to 20 m.

3. Paleomagnetism

Individual specimens, 25 mm in diameter and 25 mm long, were prepared from samples in the laboratory. Natural remanent magnetization (NRM) was measured with a Schonstedt SSM-1A spinner magnetometer. All samples were magnetically cleaned by thermal demagnetization, because this is more suitable technique than alternating field demagnetization for erasing the secondary magnetic component in welded tuffs [15,16]. Specimens were heated in air in a non-inductively wound electric furnace enclosed in three cylindrical mu-metal shields, and the specimens were cooled in the three cylindrical mu-metal shields after extraction of the electric furnace. The residual magnetic field in which the specimens were cooled was less than 10 nT.

3.1. Identification of primary component

Three pilot specimens from each site were progressively demagnetized in steps of 100°C or 150°C to the Curie temperature with increment steps reduced to 30°C or 50°C near the Curie temperature (Fig. 2). More than 70% of the welded tuffs were found to have single-component magnetizations. The change in NRM directions of pilot specimens was less than 10° (Fig. 2A). One specimen from all independently oriented samples for the site with a single-component magnetization

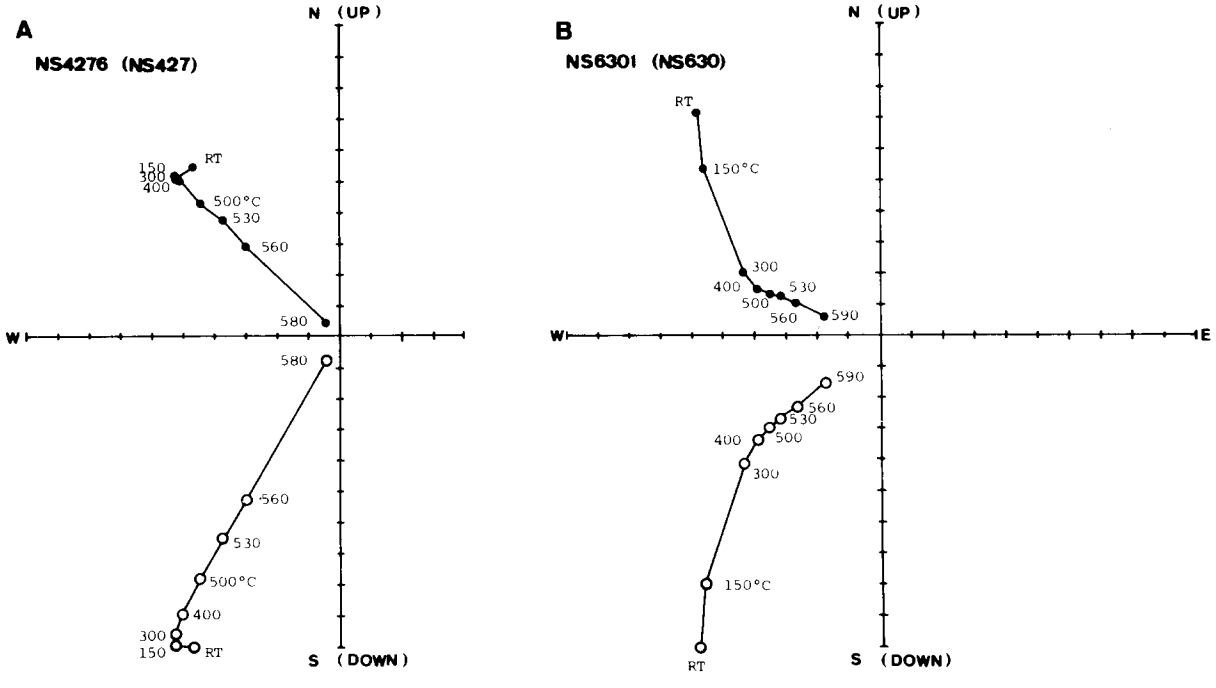


Fig. 2. Zijderveld component plot for a sample with a single-component magnetization NS4276 (A) of site NS427 and that with a multi-component magnetization NS6301 (B) of site NS630 during stepwise thermal demagnetization. Solid (open) symbols are the projection on the horizontal (vertical) plane. In (A) change in direction is less than ten degrees. In (B) some welded tuffs and granodiorites show a large change in direction up to 400°C accompanying removal of a low-blocking temperature component of present field direction, and the magnetization vector then decays more or less uniformly toward the origin. The stable end points produced during demagnetization runs are accepted to be the characteristic remanent directions for samples with multi-component magnetizations.

TABLE 1
Summary of paleomagnetic results from Northeast Japan

Area	Age (Ma)	Site	Demagnetization (°C)	N	D(°)	I(°)	α_{95}(°)	k	Rock type	Locality
(1) Ninohe										
Nisatai Fm.	32(F-T) [g]	NS630	500 B	10	−67.2	39.8	2.3	446.9	dacite welded tuff	141°20.0′, 40°17.3′
(2) Taihezan										
Okuramata Fm.	14.1(K-A) [e]	NS413	400 A	8	−10.6	42.9	4.7	140.9	rhyolite welded tuff	140°23.3′, 39°43.1′
		NS417	400 B	10	−9.9	61.1	5.1	92.2	rhyolite welded tuff	140°23.8′, 39° 42.5′
(3) Oga Peninsula										
Daijima Fm.										
Hokakejima dacite	21.5 [c], 20 [b]	NS311	400 B	6	−29.1	57.1	2.6	643.7	dacite welded tuff	139°45.8′, 39°51.4′
	(F-T)	NS633	560 B	11	−31.6	40.9	3.6	163.3	dacite welded tuff	139°45.8′, 39°51.3′
Akashima Fm.										
Nyudozaki igneous		NS302	400 A	11	−13.0	47.3	3.3	196.1	dacite welded tuff	139°42.2′, 40°00.2′
rocks	51.4 (F-T) [c]				(10.5	64.7)*				
		NS303	400 A	11	161.3	54.1	2.5	338.1	dacite welded tuff	139°41.9′, 39°59.6′
					(−5.3	62.6)*				
		NS304	400 A	11	−14.7	76.6	2.6	313.1	dacite welded tuff	139°42.4′, 40°00.2′
					(4.3	63.1)*				
		NS305	400 A	12	−17.1	28.4	4.9	79.3	dacite welded tuff	139°42.1′, 40°00.0′
					(−5.0	59.3)*				
(4) Yuzawa										
Hatamura Fm.	> 22, < 25(K-A) [d]									
Zentoku Tuff		NS512	300 A	10	132.7	−33.8	3.6	178.9	rhyolite welded tuff	140°19.1′, 39°15.7′
Masikozawa Tuff		NS513	300 A	11	134.9	−73.6	3.5	174.6	pyroxene andesite welded tuff	140°19.9′, 39°11.0′
		NS515	300 A	6	146.5	−74.1	7.7	77.4	pyroxene andesite welded tuff	140°20.0′, 39°10.9′
		NS516	300 A	8	140.0	−76.1	3.6	239.7	pyroxene andesite welded tuff	140°19.6′, 39°11.2′
Okatsugawa Fm.										
Minamisawa welded tuff	25(K-A) [d]	NS430	300 A	10	127.4	−48.4	2.7	323.3	dacite welded tuff	140°22.1′, 39°02.1′
		NS520	300 B	8	178.2	−75.1	10.9	26.9	dacite welded tuff	140°21.7′, 39°01.6′
		NS522(A)	300 A	6	133.2	−41.4	7.4	82.9	dacite welded tuff	140°21.9′, 39°01.9′
		NS522(B)	300 A	9	138.6	−50.8	4.2	149.8	dacite welded tuff	140°21.9′, 39°01.9′
		NS525	500 B	8	143.1	−54.7	8.1	48.1	dacite welded tuff	140°21.3′, 39°00.9′
		NS527	300 A	12	142.2	−53.1	2.0	450.4	dacite welded tuff	140°21.6′, 39°00.3′
Yunosawagawa Fm.	> 25(K-A) [d]	NS426	300 A	9	−56.1	67.9	2.7	361.2	pyroxene andesite welded tuff	140°22.1′, 39°06.7′
		NS427	300 A	9	−41.8	48.5	1.9	713.4	pyroxene andesite welded tuff	140°21.9′, 39°07.0′
		NS518	300 A	10	−31.8	72.9	2.7	330.3	pyroxene andesite welded tuff	140°24.6′, 39°01.4′

(5) Zao										
Yamadera Fm.										
Narasawa welded tuff	11.3(K-A) [d]	NS609	400 A	12	165.5	−66.9	3.3	175.4	dacite welded tuff	140°25.2′, 38°20.7′
		NS611	400 A	11	166.5	−71.8	1.9	608.0	dacite welded tuff	140°25.4′, 38°20.3′
(6) Northern Asahi										
Higashi-Ohotorigawa Fm.	21(F-T) [e]	NS317	400 A	12	152.5	−50.5	2.4	322.1	pyroxene andesite welded tuff	139°46.5′, 38°28.1′
Ohari granodiorite	59(K-A) [a]	NS318	400 A	12	−19.5	57.7	3.5	153.6	granodiorite	139°47.9′, 38°32.2′
		NS319	400 A	10	−29.2	25.2	4.4	122.3	granodiorite	139°48.0′, 38°32.4′
Suganodai granodiorite	61(K-A) [a]	NS320	300–560 B	11	118.1	−47.7	8.7	28.6	granodiorite	139°42.3, 38°38.2′
(7) Western Asahi										
Tenjyoyama Fm.	>15, <34(K-A) [d]	NS118	500 A	7	52.4	48.9	3.6	284.2	dacite welded tuff	139°33.8′, 38°23.0′
Kitaoguni Fm.	34.4(K-A) [d]									
	19(F-T) [e]	NS213	500 A	12	179.3	−56.7	2.6	274.3	rhyolite welded tuff	139°40.6′, 38°21.4′
	22(F-T) [e]	NS215	300 A	9	−176.1	−57.9	2.5	433.5	rhyolite welded tuff	139°40.1′, 38°20.4′
	23(F-T) [e]	NS219	400 A	9	−157.6	−59.4	4.3	141.1	rhyolite welded tuff	139°42.4′, 38°15.8′
		NS220	500 B	9	−138.1	−61.4	5.3	94.7	rhyolite welded tuff	139°42.4′, 38°15.9′
		NS221	500 B	9	−116.3	−81.0	3.5	222.8	rhyolite welded tuff	139°41.9′, 38°15.8′
Budo Fm.		NS212	300 A	10	−140.6	−54.0	4.2	136.3	rhyolite welded tuff	139°34.8′, 38°20.1′
Asahi Rhyolite	50.3, 54.2	NS119	500 B	7	60.0	40.3	5.9	107.0	rhyolite welded tuff	139°35.1′, 38°18.8′
	48.1(K-A) [e]	NS322	500 B	12	38.8	53.8	6.1	51.3	rhyolite welded tuff	139°35.0′, 38°18.8′
(8) Iide										
Deyu Granodiorite	92(K-A) [a]	NS111	400 A	11	−58.6	5.05	5.1	80.0	granodiorite	139°19.6′, 37°49.0′
(9) Ashio										
Matsukizawa granodiorite	116(K-A) [a]	NS405	400 A	10	−10.7	61.9	4.9	99.1	granodiorite	139°25.3′, 36°41.4′
		NS407	400 A	6	−34.1	58.4	5.6	143.2	granodiorite	139°25.2′, 36°41.5′

All data except for those of granodiorite are after tilt correction. The ages are data from: [a] Kawano and Ueda [34], [b] Nishimura and Ishida [35], [c] Suzuki [36], [d] Konda and Ueda [37], [e] Matsuda et al. [38], [f] MMAJ [30] and [g] Samata [24]. (K-A) = K-Ar dating, (F-T) = fission track dating. Demagnetization: the magnetic cleaning temperature and behaviour of NRM direction during thermal demagnetization; A = single-component magnetization, change in NRM direction was less than 10° during demagnetization; B = multi-component magnetization, change was more than 10°. N is the number of samples in site mean computation, D is the site mean declination, I is the site mean inclination, α_{95} is the radius of the 95% circle of confidence around the site mean direction and k is Fisher precision parameter. D and I denoted by an asterisk (*) are before tilt correction.

was demagnetized at the optimum demagnetization temperature, i.e., that which produced minimum dispersion for NRM directions from the three pilot specimens. Although the granitic rocks and some of the welded tuffs showed a large change in their NRM directions during progressive demagnetization at lower temperatures (Fig. 2B), a few rocks yielded consistent magnetic direction at high temperature (400–500°C). All specimens for these sites with multi-component magnetizations were progressively demagnetized. The stable end points during demagnetization runs were identified to be their characteristic directions.

The characteristic direction after thermal demagnetization for each site was accepted as reliable provided the following criteria were satisfied: An acceptable site has at least five specimens cut from independently oriented samples and a mean direction from the site has an error circle of 95% confidence less than 15°.

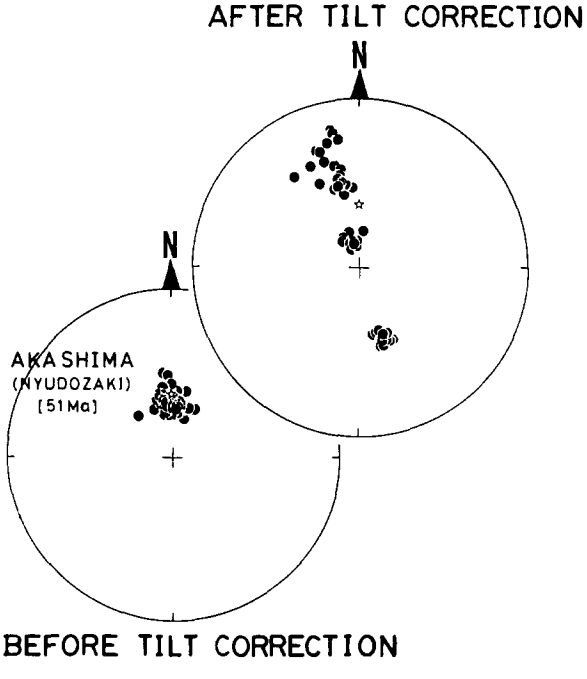


Fig. 3. Natural remanent magnetizations (NRM) (Table 1) of Nyudozaki igneous rocks in Oga Peninsula before and after application of the tilting correction. Present axial dipole field direction is shown by star. The NRM directions before bedding corrections are identical to the present axial dipole field direction. Projections are equal area, solid circles on the lower hemisphere, open circles on the upper hemisphere.

The tilting test was used to detect the secondary directions in the reliable characteristic directions (see Fig. 3). The NRMs of Nyudozaki igneous rocks (51 Ma) from Oga Peninsula were judged to be secondary remanence acquired after the peninsula was faulted and folded (Fig. 3). Their directions before tilt correction are tightly grouped and are parallel to the present axial dipole field. The angular dispersion of the NRM directions of the Nyudozaki igneous rocks after tilt correction based on the eutaxitic structure increases significantly, as is shown by a decrease of the Fisher precision parameter k [39] from 376.4, for in situ results, to 3.9 for the tilt corrected results. No further paleomagnetic investigation was conducted for those sites whose characteristic direction before tilt correction was parallel to the present axial dipole field direction.

3.2. Paleomagnetic directions

Thirty six sites were found to have a reliable primary magnetic component through thermal demagnetization and tilting test (Table 1): six sites are from Cretaceous/Paleogene granitic rocks, twenty six sites are from Paleogene/early Miocene igneous rocks and four sites are from middle Miocene rocks.

Cretaceous to Paleogene granitic rocks (110–59 Ma). Declinations of the Cretaceous granite from the southernmost part of Northeast Japan (Iide and Ashio areas) show a westward deflection which is consistent with the declination value previously reported from Northeast Japan of the Cretaceous period [12,40,41].

This westward deflection is also observed for Paleogene granitic rocks from Uetsu area. We measured the same granodiorites (NS320; Suganodai) whose NRM had been reported to have easterly declination, i.e. $D = 45°$ [12]. Although the NRM directions of the rocks were dispersed before thermal demagnetization, they were roughly directed eastward (Fig. 4). The NRM after thermal demagnetization above 400°C, however, changed its polarity and grouped at about $D = 120°$, $I = -50°$. The negative inclination indicates reversed polarity, and thus the declination for normal polarity should be $-60°$. Other rocks from Uetsu area also showed a westerly direction

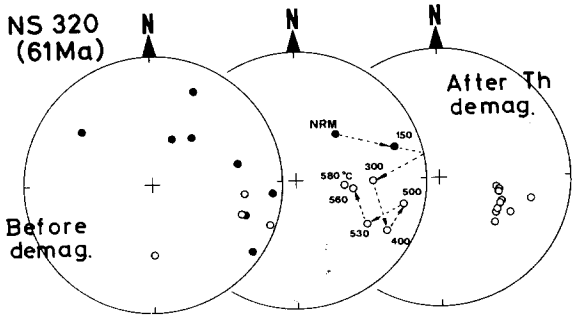


Fig. 4. Thermal demagnetization of specimens from Suganodai granodiorite with K-Ar age of 61 Ma. Cleaned direction (right) of natural remanent magnetization (NRM) together with NRM directions before demagnetization (left) and example of progressive thermal demagnetization (center). The scatter of directions before demagnetization is reduced by thermal demagnetization technique, and the clustered direction with reversed polarity ($D = 118.1°$, $I = -47.7°$) is identified. The NRM direction of Suganodai diorite was reported to be $D = 45°$ and $I = 34°$ by Kawai et al. [12]. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

with normal polarity. The westward deflection is probably a more realistic estimate of the declination value for these granitic rocks with age of 60 Ma. Eastward deflections that were previously reported could be due to insufficient demagnetization.

Paleogene to early Miocene welded tuffs (54–19 Ma). The distribution of sampling sites for Paleogene welded tuffs covers a large area from 38.3° to 40.3°N in latitude. Almost all welded tuffs except for those from the southern area (Western Asahi) show a westward deflection in declination value. The declinations lie between $-2°$ and $-67°$. The reliability of the westerly declination is ascertained through the presence of normal and reversed polarities (Fig. 5). The declination value of the Paleogene to early Miocene period is also char-

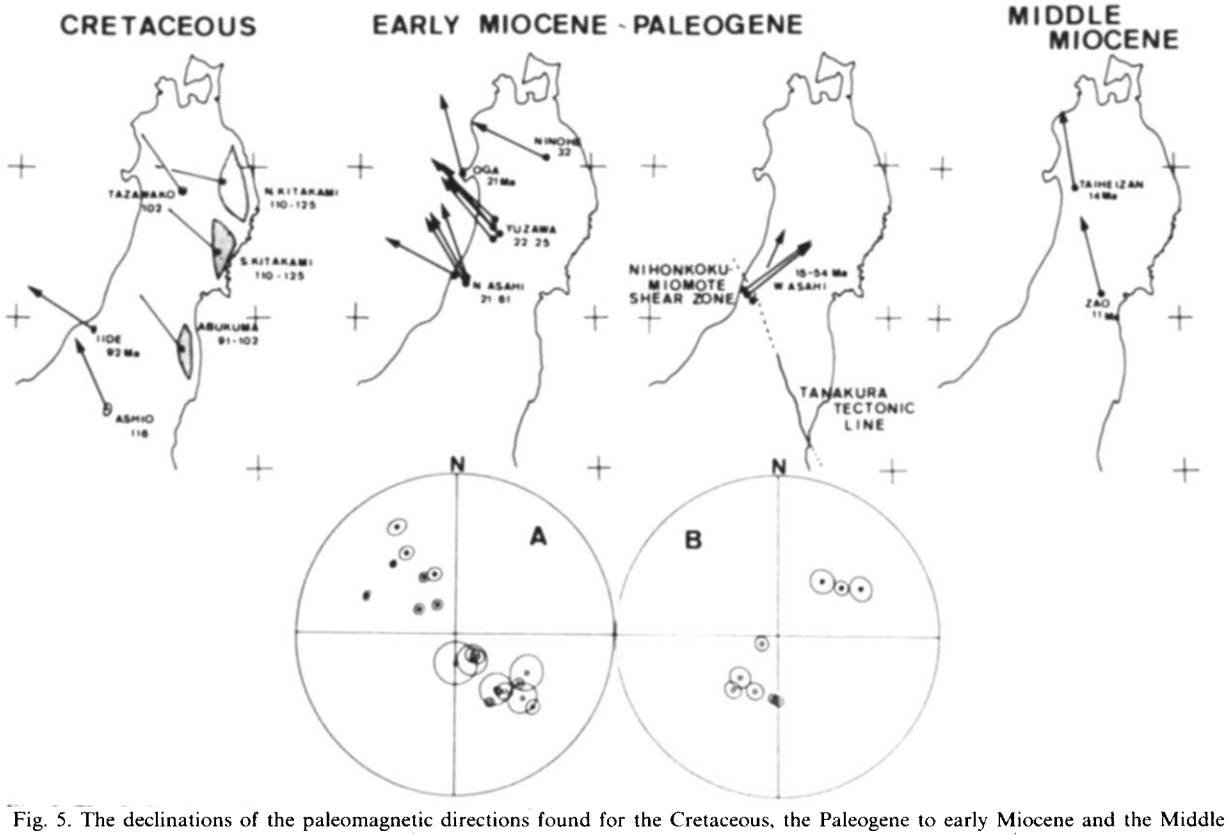


Fig. 5. The declinations of the paleomagnetic directions found for the Cretaceous, the Paleogene to early Miocene and the Middle Miocene stages of Northeast Japan. The Paleogene to early Miocene data of Northeast Japan are divided into those of the Western Asahi and other areas: Paleomagnetic field directions and 95% confidence circles of sites are also shown for the Western Asahi (B) and other areas (A). Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere. The Cretaceous data from North and South Kitakami and Tazawako are after Ito and Tokieda [40], Ito et al. [41], and those from Abukuma are after Kawai et al. [12].

acterized by the westerly declination for a large part of Northeast Japan.

In contrast with the westward deflection, an eastward deflection has been discovered from Paleogene welded tuffs of Western Asahi. The reliability of the easterly direction is ascertained through the presence of normal and reversed polarities (Fig. 5). The easterly direction of this area is definitely distinguished from the westerly direction observed in the other areas of Northeast Japan at 95% confidence level ($D = 35.9°$, $I = 59.8°$, $\alpha_{95} = 10.4°$ vs. $D = -41.2°$, $I = 56.5°$, $\alpha_{95} = 7.2°$). Since the easterly declination has been found in rocks with age ranging from 54 to about 15 Ma, the easterly declination is not probably attributed to the short time phenomena of geomagnetic field such as the field transition or excursion.

Middle Miocene (11 and 14 Ma). Miocene welded tuffs were collected from the Taiheizan and Zao areas. A mean NRM direction from both areas is equivalent to the present axial dipole field within 95% confidence limits after tilt correction (see Table 2). The remanent magnetizations of the Middle Miocene welded tuffs are reasonably accepted to be a primary component, because the northerly directions are obtained after tilt correction.

4. Discussion

4.1. Counter-clockwise rotation of Northeast Japan

Directions of the declinations for three periods, the Cretaceous, the Paleogene to early Miocene, and Middle Miocene, are drawn in Fig. 5. The Cretaceous paleomagnetic data of this work combined with previous data [12,40,41] indicate that Northeast Japan of the Cretaceous period is characterized by a westerly deflection in declination. Westerly directions have been also observed through remeasurements of Paleogene granitic rocks, and discovered from Paleogene to early Miocene welded tuffs. Because well-grouped westerly directions have been obtained after tilt correction for Tertiary welded tuffs, the westerly deflection in declination possibly indicates true tectonic rotation about a vertical axis [42]. These results imply that the counter-clockwise rotation of Northeast Japan occurred during late Tertiary. On the basis of the Cretaceous data (Fig. 5), the rotated block is concluded to extend southward to Ashio area.

TABLE 2
Average paleomagnetic directions from Late Tertiary volcanic rocks

Formation name	N	$\bar{D}(°)$	$\bar{I}(°)$	$\alpha_{95}(°)$	k	VGP	$\partial p(°)$	$\partial m(°)$
11–14 Ma								
Yamadera	2 R	−14.5	69.7	–	–			
Okuramata	2 N	−10.1	51.1	–	–			
Mean (sites)	4	−11.7	60.5	15.3	37.2	70.2°N, 80.7°E	17.7	23.3
(formations)	2	−11.7	60.4	–	–	69.5°N, 80.8°E	–	–
20–32 Ma								
Higashiohtorigawa	1 R	−27.4	50.5	–	–			
Daijima	2 N	−30.0	47.6	–	–			
Hatamura	4 R	−43.4	64.5	24.2	15.4			
Okatsugawa	6 R	−40.5	54.1	11.6	34.4			
Yunosawagawa	3 N	−43.5	63.1	21.4	34.2			
Nisatai	1 N	−66.9	39.4	–	–			
Mean (sites)	17	−41.2	56.5	7.2	25.7	60.8°N, 57.8°E	7.5	10.4
(formations)	6	−42.5	54.0	11.0	37.8	57.0°N, 56.0°E	10.8	15.4

Directions are presented as the value at the representative point of Northeast Japan (141°E, 39°N). N is number of sites, $\bar{D}$, $\bar{I}$ are the mean declination and inclination, α_{95} and k are Fisher statistic parameters, and ∂p and ∂m are semiaxes of the oval of 95% confidence around the pole. The mean directions for each stage, 11–14 Ma and 20–32 Ma, are calculated by assigning unit weight to formations and sites, respectively.

Easterly declination values in the Paleogene obtained from southern part of Northeast Japan (Western Asahi) are discordant with the data from the rest of Northeast Japan (Fig. 5). The area is expected to have undergone a different block movement from the other areas later than 15 Ma. The easterly declination data could be explained by the clockwise rotation of local block due to dextral strike slip faulting [42]. Sampling sites are situated along the Nihonkoku-Miomote shear zone which is the northwestward extension of the Tanakura tectonic line [31]: the distance between the sampling points and the shear zone is less than 9 km. On the basis of the geological observation [43,44], right-lateral displacement is expected to have occurred along the Nihonkoku-Miomote shear zone, and the vertical and small-scale right-lateral movements also occurred along the Tanakura Tectonic Line during the Miocene. The timing of the clockwise rotation of this area is consistent with the period of activity of the Tanakura Tectonic Line. The Paleogene igneous rocks of this area probably record only the local block movement, and tectonic rotation of Northeast Japan may be concealed by this movement.

4.2. Timing and amount of the rotation of Northeast Japan

We can estimate the timing of the counter-clockwise rotation from the paleomagnetic data combined with geochronological data of Northeast Japan except for Western Asahi. The declination value is drawn as a function of age in Fig. 6. K-Ar age determination assigns 14 Ma and 11 Ma for the welded tuffs of the Okuramata (NS 413, NS417) and Yamadera (NS609, NS611) Formations [37,38] whose NRMs are northward. The youngest rocks with westerly declination value, the Hokakejima dacite (NS311, NS633) and the Higashi-Ohtorigawa Formation (NS317), showed a fission track age of 21 Ma [35,36,38]. Thus, the timing of the rotation is presumed to be from 21 Ma to between 14 and 11 Ma.

The amount of rotation is evaluated through statistical calculation of the data of rocks between 32 Ma and 21 Ma except for those from Western Asahi (Table 2). The mean direction is $D = -41.2°$, $I = 56.5°$ and $\alpha_{95} = 7.2°$: The value is that at the representative point of Northeast Japan (141°E, 39°N). Taking into account of the effect

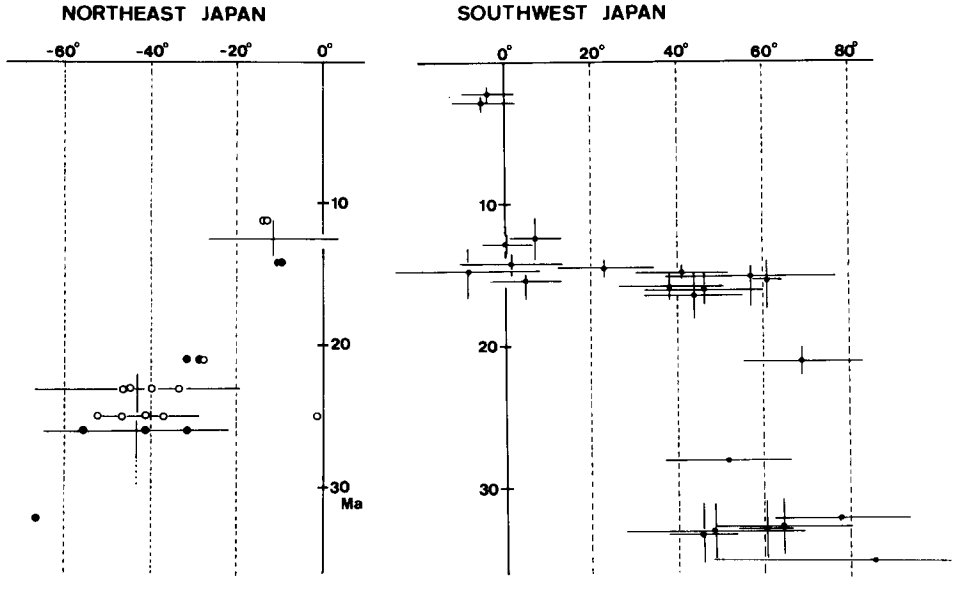


Fig. 6. Declination versus age for Northeast Japan and Southwest Japan. Northeast Japan: both mean declination data of sites and rock units are shown. The declination error bars of rock units means are the α_{95}. Closed circles show the data with normal polarity, open circles show the data with reversed polarity. Southwest Japan: only mean declination data of rock units are shown after Otofuji et al. [18].

of apparent polar wander since 20 Ma, the amount of rotation and its uncertainty are $R = 46.8°$ and $\Delta R = 14.1°$ (the R and ΔR values are after Margill et al. [45]): we used the paleofield direction and confidence oval at the representative point of Northeast Japan calculated from the 20 Ma paleopole (85°N, 198°E, $\alpha_{95} = 4°$) of Northern Eurasia [46].

Difference between the observed inclination value and the calculated value from the apparent polar wonder path is $-4.1 \pm 9.1°$ for the period between 30 Ma and 20 Ma, which indicates no significant latitudinal translation for Northeast Japan relative to Northern Eurasia. Comparing the inclination value for Northeast Japan at the period between 35 and 20 Ma with that for Southwest Japan [16] ($I = 56.5°$, $\alpha_{95} = 7.2°$ vs. $I = 48.6°$, $\alpha_{95} = 6.5°$), Northeast Japan, as it is at present, appears to have been situated at higher latitude than Southwest Japan. It is concluded that Northeast Japan has been subjected to the counterclockwise rotation through $47 \pm 14°$ since 21 Ma without suffering significant north-south translation.

The timing of the rotation of Northeast Japan is compared with that of Southwest Japan (Fig. 6). Paleomagnetic work combined with fission-track dating for welded tuffs from San'in district of Southwest Japan showed that the clockwise rotation of Southwest Japan took place between 21 Ma and 12 Ma [16]. The magnetostratigraphy for the Setouchi Miocene Series of eastern Southwest Japan suggests about 16 Ma for the start of rotation [17]. The best curve fitting analysis for rotation versus age for all the reliable data from Southwest Japan determines that the climax of the rotation was at 14.9 Ma [18]. Although the paleomagnetic data from Northeast Japan are much smaller in number than those from Southwest Japan to estimate the timing of rotation, the timing of rotation of Northeast Japan (21 Ma to between 14 and 11 Mà) is comparable to that of Southwest Japan. Both Northeast Japan and Southwest Japan probably underwent rotation with respect to the Asian continent at almost the same period in middle Miocene.

4.3. Model of the rotation of Northeast Japan and Southwest Japan

We propose the following model to explain the amount of rotation for both Northeast Japan and Southwest Japan (see Fig. 7):

(1) The rotation of both arcs were phenomena associated with the rifting process of Japan arc due to back arc opening of the Japan Sea.

(2) The differential rotation about different pivots occurred between Northeast Japan and Southwest Japan.

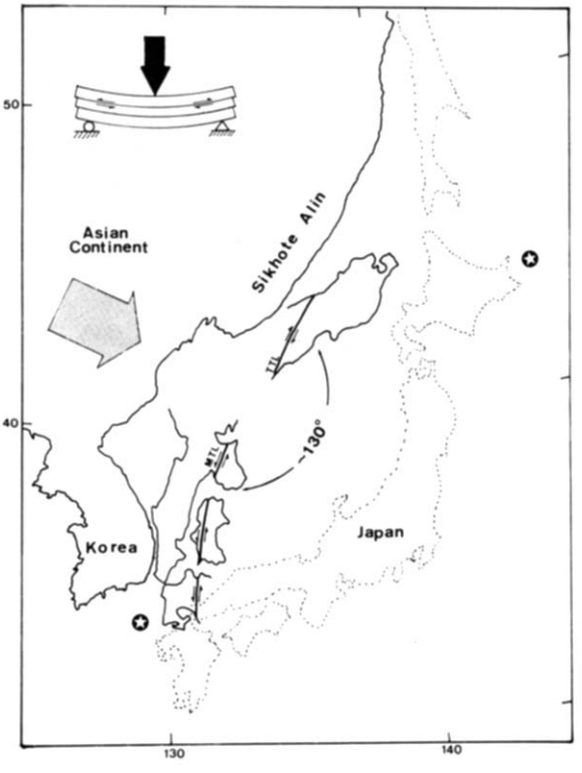


Fig. 7. Rotation model of Northeast and Southwest Japan. Position of the primary blocks of Northeast and Southwest Japan before 21 Ma and present position of the Japan arc (dashed coast lines). The rotation pivots (146°E, 44°N) for Northeast Japan and (129°E, 34°N) for Southwest Japan are shown by stars. The axis of Northeast Japan arc makes an angle of about 130° with that of Southwest Japan arc before both blocks are rifted from the Asian continental margin. The inset shows a piled up planks model: If separate planks are not fastened together, sliding at the surfaces of their contact will take place. Right lateral movement is expected along the Tanakura Tectonic Line (*TTL*) and left lateral along the Median Tectonic Line (*MTL*).

(3) Northeast Japan rotated about a northern pivot (146°E, 44°N) through 50°, while Southwest Japan rotated about a southern pivot (129°E, 34°N) through 54°. The paleoposition of the Japan arc is estimated on the basis of a paleomagnetic constraint and a physiographycal assumption that the margins of the rifted arcs are fitted to continental margins prior to drift (the isobath of 3000 m is chosen for a continental margin of the northern part of the Japan Sea).

(4) The differential rotation occurred during the period between 21 Ma and 12 Ma. The climax of the rotation is at about 15 Ma.

Our rotation model shows that the axis of the Northeast Japan arc formed an angle of about 130° with that of Southwest Japan (< -shape) before 21 Ma. The reconstruction for paleoshape of the Japan arc (< -shape) depends on the paleomagnetic constraint that the declinations of the period between 20 and 35 Ma from both arcs should be parallel to paleo-meridian. The previous models [12,14] based on the paleomagnetic data obtained before 1980 suggested that the Japan arc had been straight in figure before both arcs underwent the differential rotation. Since our study and the recent paleomagnetic works, in which the data are obtained after the treatments of thermal demagnetization and tilt correction, have produced accurate estimate of the amount of rotation for both Northeast and Southwest Japan arcs, a < -shape is a more realistic estimate for the paleoshape of the Japan arc than the straight figure. This inference is supported by geometrical fitting: the < -shape of Japan arc is fairly similar to the shape of the coast line from Sikhote Alin to the Korean Peninsula.

We delineate the process for rotation of both parts of Japanese islands as follows: The eastern margin of the Asian continent was subjected to a force (indicated by a large arrow in Fig. 7) exerted either from the eastward flow of the asthenosphere with respect to the continental lithosphere [47,48] or the tectonic flow induced by downgoing slab [49] at the Miocene period. Since both Northeast Japan and Southwest Japan at the continental margin had thinner lithosphere than the Korean Peninsula and China [50–52], the lithosphere below both parts of Japan was not strong enough to resist the active extension. Both parts, therefore, rifted from the continent. Southwest Japan rotated clockwise and Northeast Japan counter-clockwise at about 15 Ma. The diamond shaped back arc basin, the Japan Sea, was created by the differential rotation of the two parts of Japan [53].

The Japan arc consists of several geologic terranes, which are zonally arranged sub-parallel to the Japan trench or parallel to the Nankai Trough [54]. When the primary block of the Japan arc at the continental margin was subjected to the force which rifted the block from the continental margin, these terranes prior to rifting probably behaved as a pile of elastic planks (see Fig. 7). Sliding at the boundaries of the terranes will take place. Right-lateral movement is expected at the Tanakura Tectonic Line in Northeast Japan and left-lateral movement at the Median Tectonic Line in Southwest Japan in the Paleogene to Miocene period. This inference explains the paleomagnetic and the geological observation [43,55] that Tanakura Tectonic Line was a right-lateral strike slip fault and the Median Tectonic Line a left-lateral strike slip fault during the Paleogene to the Miocene.

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